

STT4890 Football Homework

Christian Etherson

2025-11-20

```
library(mgcv)

## Loading required package: nlme
## This is mgcv 1.9-3. For overview type 'help("mgcv-package")'.
library(nflfastR)
library(dplyr)

##
## Attaching package: 'dplyr'
## The following object is masked from 'package:nlme':
##
##      collapse
## The following objects are masked from 'package:stats':
##
##      filter, lag
## The following objects are masked from 'package:base':
##
##      intersect, setdiff, setequal, union
library(ggplot2)
```

Loading Data

```
nflData <- load_pbp(seasons = 2025)
nflData

## -- nflverse play by play data -----
## i Data updated: 2025-11-23 20:53:33 EST
## # A tibble: 30,263 x 372
##   play_id game_id      old_game_id home_team away_team season_type week posteam
##   <dbl> <chr>      <chr>      <chr>      <chr>      <chr>      <int> <chr>
## 1      1 2025_01_AR~ 2025090705 NO        ARI        REG          1 <NA>
## 2     40 2025_01_AR~ 2025090705 NO        ARI        REG          1 ARI
## 3     63 2025_01_AR~ 2025090705 NO        ARI        REG          1 ARI
## 4     85 2025_01_AR~ 2025090705 NO        ARI        REG          1 ARI
## 5    115 2025_01_AR~ 2025090705 NO        ARI        REG          1 ARI
## 6    135 2025_01_AR~ 2025090705 NO        ARI        REG          1 ARI
## 7    166 2025_01_AR~ 2025090705 NO        ARI        REG          1 ARI
## 8    188 2025_01_AR~ 2025090705 NO        ARI        REG          1 ARI
```

```
## 9      214 2025_01_AR~ 2025090705 NO      ARI      REG      1 NO
## 10     243 2025_01_AR~ 2025090705 NO      ARI      REG      1 NO
## # i 30,253 more rows
## # i 364 more variables: posteam_type <chr>, defteam <chr>, side_of_field <chr>,
## #   yardline_100 <dbl>, game_date <chr>, quarter_seconds_remaining <dbl>,
## #   half_seconds_remaining <dbl>, game_seconds_remaining <dbl>,
## #   game_half <chr>, quarter_end <dbl>, drive <dbl>, sp <dbl>, qtr <dbl>,
## #   down <dbl>, goal_to_go <int>, time <chr>, yrdln <chr>, ydstogo <dbl>,
## #   ydsnet <dbl>, desc <chr>, play_type <chr>, yards_gained <dbl>, ...
```

Filtering for Fourth Down Plays and Selecting 15 Games

```
set.seed(1234)

nflDataFourth <- nflData %>%
  distinct(game_id) %>%
  slice_sample(n = 15)

nflDataFourth <- nflData %>%
  semi_join(nflDataFourth, by = "game_id") %>%
  filter(down == 4,
         play_type != "no_play")
nflDataFourth

## -- nflverse play by play data -----
## i Data updated: 2025-11-23 20:53:33 EST
## # A tibble: 214 x 372
##   play_id game_id      old_game_id home_team away_team season_type week posteam
##   <dbl> <chr>      <chr>      <chr>      <chr>      <chr>      <int> <chr>
## 1     283 2025_02_PH~ 2025091411 KC        PHI        REG          2 PHI
## 2     370 2025_02_PH~ 2025091411 KC        PHI        REG          2 KC
## 3     461 2025_02_PH~ 2025091411 KC        PHI        REG          2 PHI
## 4     679 2025_02_PH~ 2025091411 KC        PHI        REG          2 KC
## 5    1126 2025_02_PH~ 2025091411 KC        PHI        REG          2 KC
## 6    1315 2025_02_PH~ 2025091411 KC        PHI        REG          2 PHI
## 7    1473 2025_02_PH~ 2025091411 KC        PHI        REG          2 KC
## 8    1982 2025_02_PH~ 2025091411 KC        PHI        REG          2 PHI
## 9    2118 2025_02_PH~ 2025091411 KC        PHI        REG          2 KC
## 10   2250 2025_02_PH~ 2025091411 KC        PHI        REG          2 PHI
## # i 204 more rows
## # i 364 more variables: posteam_type <chr>, defteam <chr>, side_of_field <chr>,
## #   yardline_100 <dbl>, game_date <chr>, quarter_seconds_remaining <dbl>,
## #   half_seconds_remaining <dbl>, game_seconds_remaining <dbl>,
## #   game_half <chr>, quarter_end <dbl>, drive <dbl>, sp <dbl>, qtr <dbl>,
## #   down <dbl>, goal_to_go <int>, time <chr>, yrdln <chr>, ydstogo <dbl>,
## #   ydsnet <dbl>, desc <chr>, play_type <chr>, yards_gained <dbl>, ...
```

For my randomly selected 15 games of the 2025 season, there have been 228 fourth down occurrences.

Question 1)

```
fourth_down <- nflDataFourth %>%
  filter(down == 4) %>%
  select(game_id, posteam, game_seconds_remaining, score_differential, down,
         yardline_100, ydstogo, play_type)
fourth_down

## -- nflverse play by play data -----
## i Data updated: 2025-11-23 20:53:33 EST
## # A tibble: 214 x 8
##   game_id posteam game_seconds_remaining score_differential down yardline_100
##   <chr>    <chr>                <dbl>                <dbl> <dbl>      <dbl>
## 1 2025_02~ PHI                    3331                    0     4         50
## 2 2025_02~ KC                     3283                    0     4         84
## 3 2025_02~ PHI                    3216                    0     4         78
## 4 2025_02~ KC                     3000                    0     4         40
## 5 2025_02~ KC                     2508                   -7     4         38
## 6 2025_02~ PHI                    2316                    4     4         76
## 7 2025_02~ KC                     2160                   -4     4         42
## 8 2025_02~ PHI                    1802                   -3     4         39
## 9 2025_02~ KC                     1680                    0     4         64
## 10 2025_02~ PHI                    1593                    0     4         33
## # i 204 more rows
## # i 2 more variables: ydstogo <dbl>, play_type <chr>

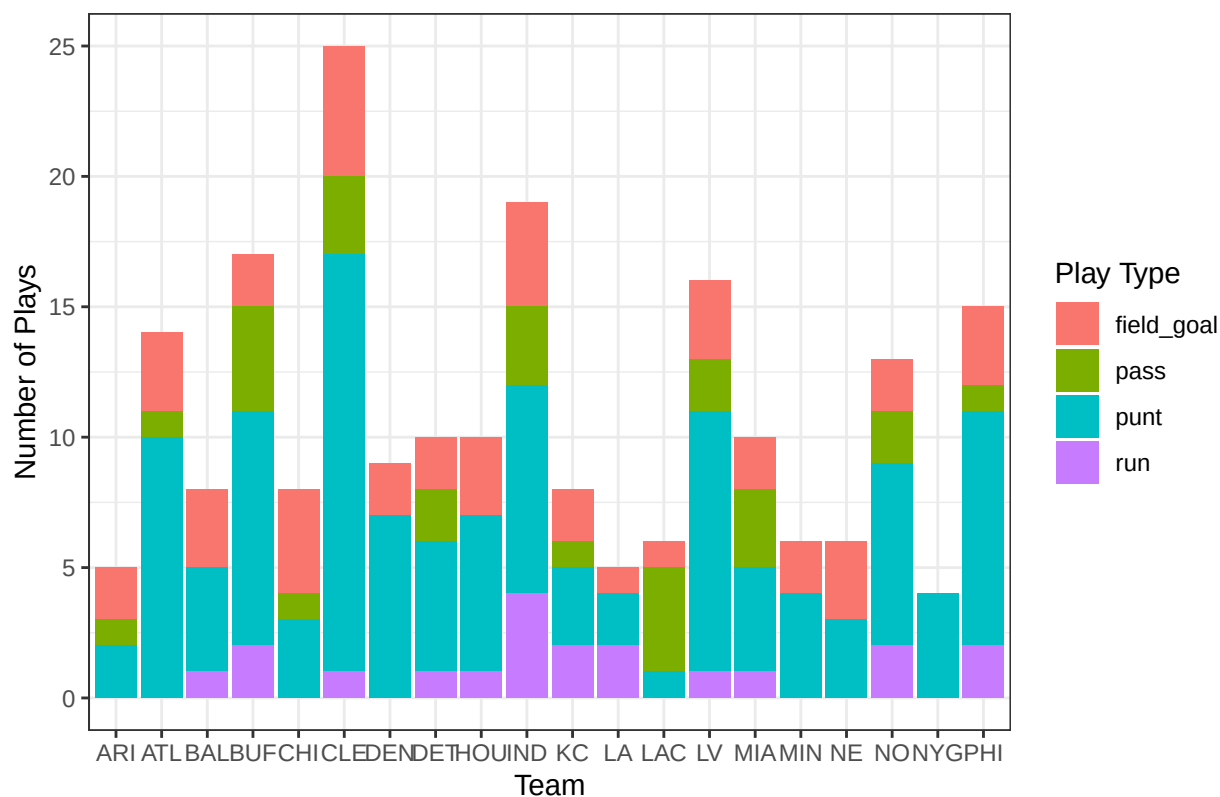
games <- unique(fourth_down$game_id)
print(games)

## [1] "2025_02_PHI_KC" "2025_05_LV_IND" "2025_06_ARI_IND" "2025_06_BUF_ATL"
## [5] "2025_06_PHI_NYG" "2025_07_IND_LAC" "2025_07_MIA_CLE" "2025_07_NO_CHI"
## [9] "2025_08_CLE_NE"  "2025_09_MIN_DET" "2025_09_NO_LA"  "2025_10_ATL_IND"
## [13] "2025_10_LV_DEN"  "2025_11_BAL_CLE" "2025_12_BUF_HOU"

ggplot(fourth_down, aes(x = posteam, fill = play_type)) +
  geom_histogram(stat = "count") +
  labs(title = "4th-Down Decisions per Team (All Games)",
       x = "Team",
       y = "Number of Plays",
       fill = "Play Type") +
  theme_bw()

## Warning in geom_histogram(stat = "count"): Ignoring unknown parameters:
## `binwidth` and `bins`
```

4th-Down Decisions per Team (All Games)



```
punt_percent <- fourth_down %>%
  group_by(posteam) %>%
  summarise(
    total_plays = n(),
    punts = sum(play_type == "punt"),
    percent_punt = (punts / total_plays) * 100
  ) %>%
  arrange(desc(percent_punt))
```

punt_percent

```
## # A tibble: 20 x 4
##   posteam total_plays punts percent_punt
##   <chr>      <int> <int>      <dbl>
## 1 NYG         4      4        100
## 2 DEN         9      7        77.8
## 3 ATL        14     10        71.4
## 4 MIN         6      4        66.7
## 5 CLE        25     16         64
## 6 LV         16     10        62.5
## 7 HOU        10      6         60
## 8 PHI        15      9         60
## 9 NO         13      7        53.8
## 10 BUF        17      9        52.9
## 11 BAL         8      4         50
## 12 DET        10      5         50
## 13 NE         6      3         50
```

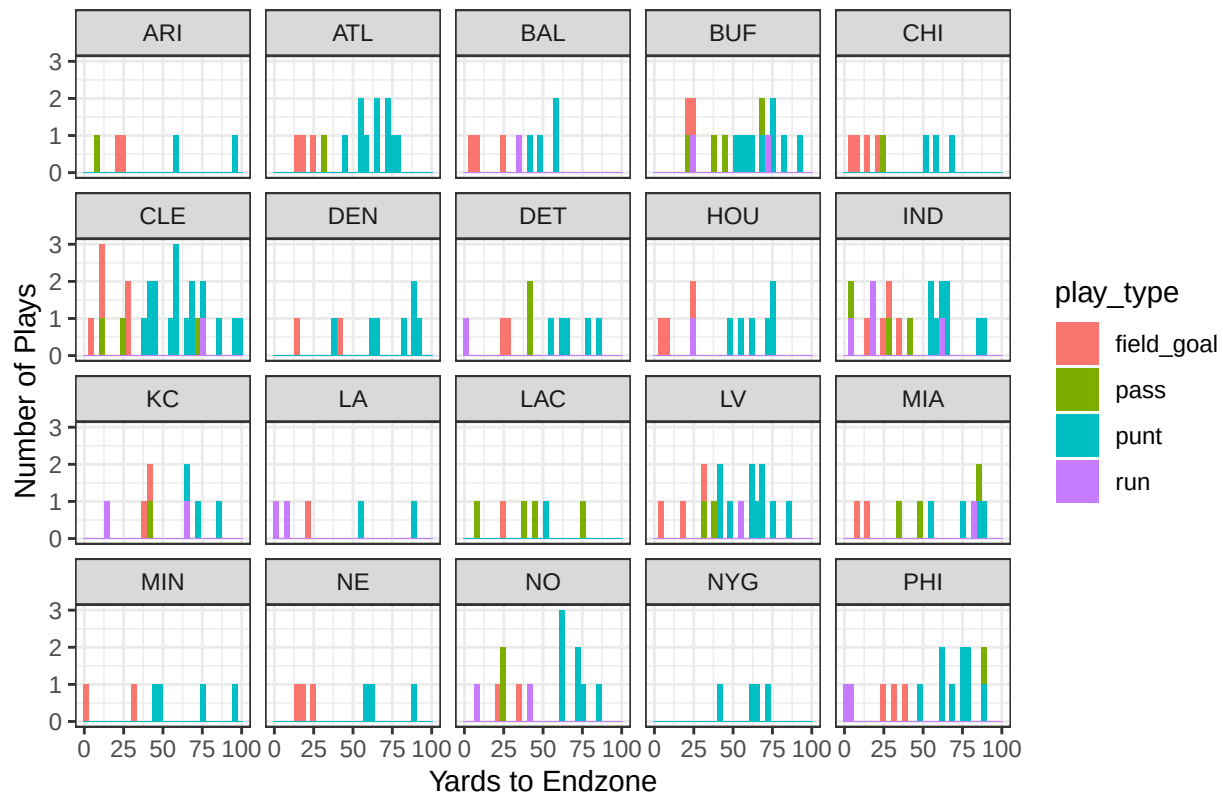
## 14 IND	19	8	42.1
## 15 ARI	5	2	40
## 16 LA	5	2	40
## 17 MIA	10	4	40
## 18 CHI	8	3	37.5
## 19 KC	8	3	37.5
## 20 LAC	6	1	16.7

From the plot and the frequency table, Buffalo punts the ball the most in fourth down situations, punting 85% of the time. The Chargers punt the least in fourth down situations, only punting 16% of the time. Cleveland, who has the highest fourth down occurrence, punted the ball 64% of the time.

```
ggplot(fourth_down, aes(x = yardline_100, fill = play_type)) +
  geom_histogram() +
  facet_wrap(~ posteam) +
  labs(title = "4th-Down Decisions by Field Position (All Games)",
       x = "Yards to Endzone",
       y = "Number of Plays") +
  theme_bw()
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.

4th-Down Decisions by Field Position (All Games)

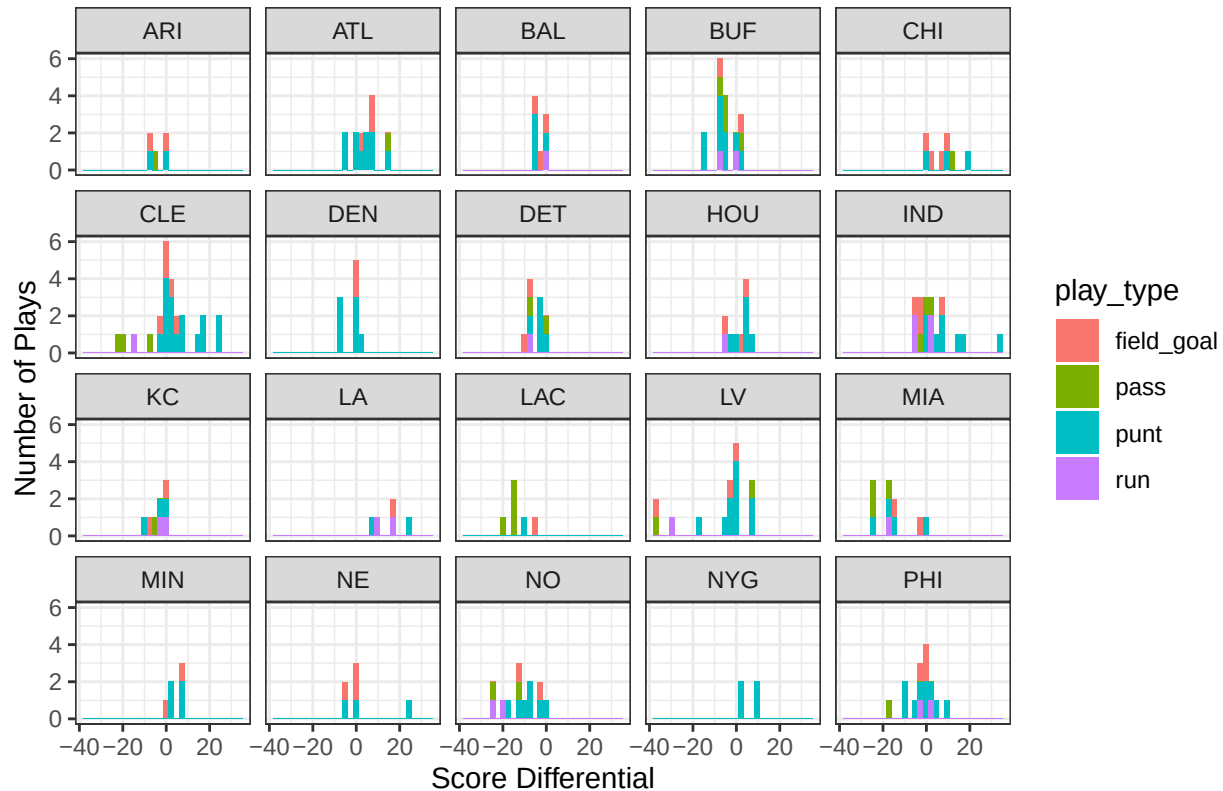


From the graph, it seems like most teams will punt when they are greater than 50 yards away from the endzone. As they get within 25 yards of the endzone, field goals become much more common. An example of this is the New York Giants, who punted the ball every time their field position was more than 50 yards from the endzone. Around 25 yards, they kicked a field goal, and in their one close yardage situation they ran the ball. A counterexample is the Miami Dolphins, who have a couple examples of going for it on fourth down well into their own territory.

```
ggplot(fourth_down, aes(x = score_differential, fill = play_type)) +
  geom_histogram() +
  facet_wrap(~ posteam) +
  labs(title = "4th-Down Decisions by Score Differential (All Games)",
       x = "Score Differential",
       y = "Number of Plays") +
  theme_bw()
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.

4th-Down Decisions by Score Differential (All Games)

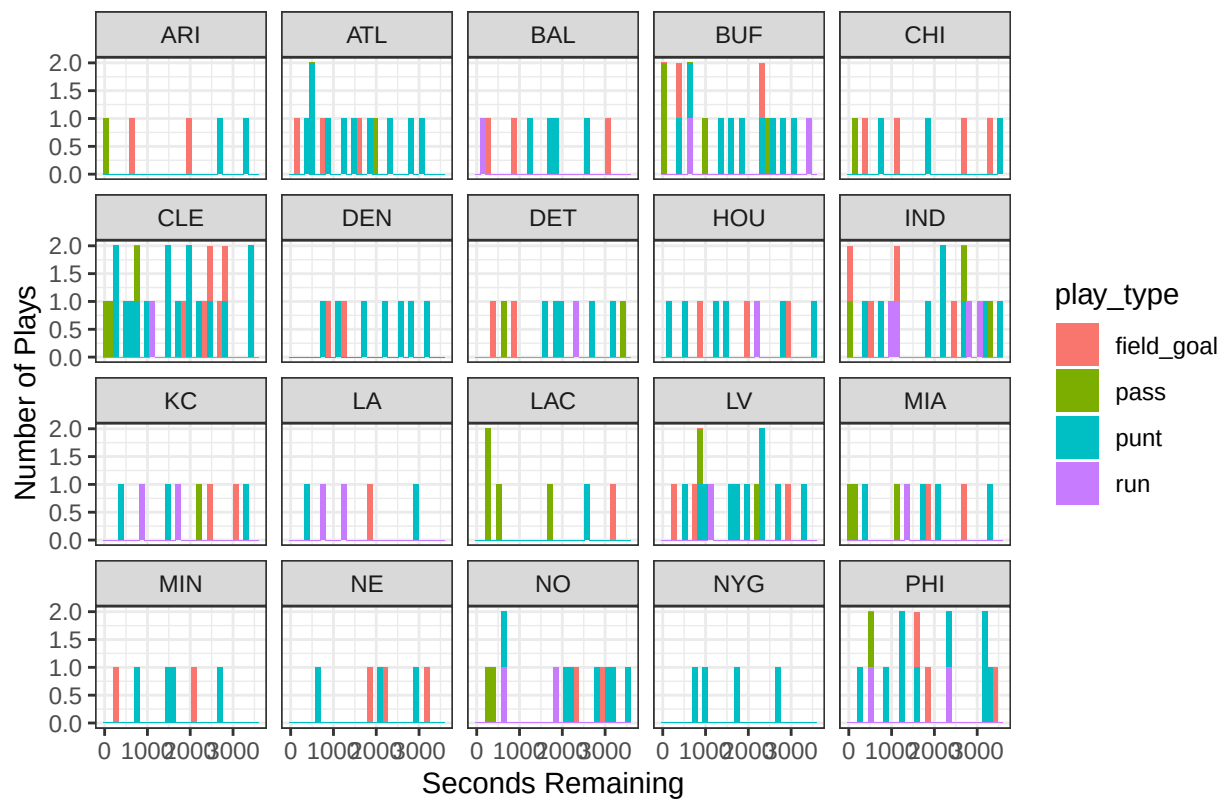


For most teams there doesn't appear to be much correlation between score differential and action on fourth down. However, when a team is more than 15 points behind, teams may be more likely to go for it. An example of this is the Cleveland Browns, who passed or ran the ball when they were behind. A counterexample is the Atlanta Falcons who punted from behind.

```
ggplot(fourth_down, aes(x = game_seconds_remaining, fill = play_type)) +
  geom_histogram() +
  facet_wrap(~ posteam) +
  labs(title = "4th-Down Decisions by Time Remaining (All Games)",
       x = "Seconds Remaining",
       y = "Number of Plays") +
  theme_bw()
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.

4th-Down Decisions by Time Remaining (All Games)

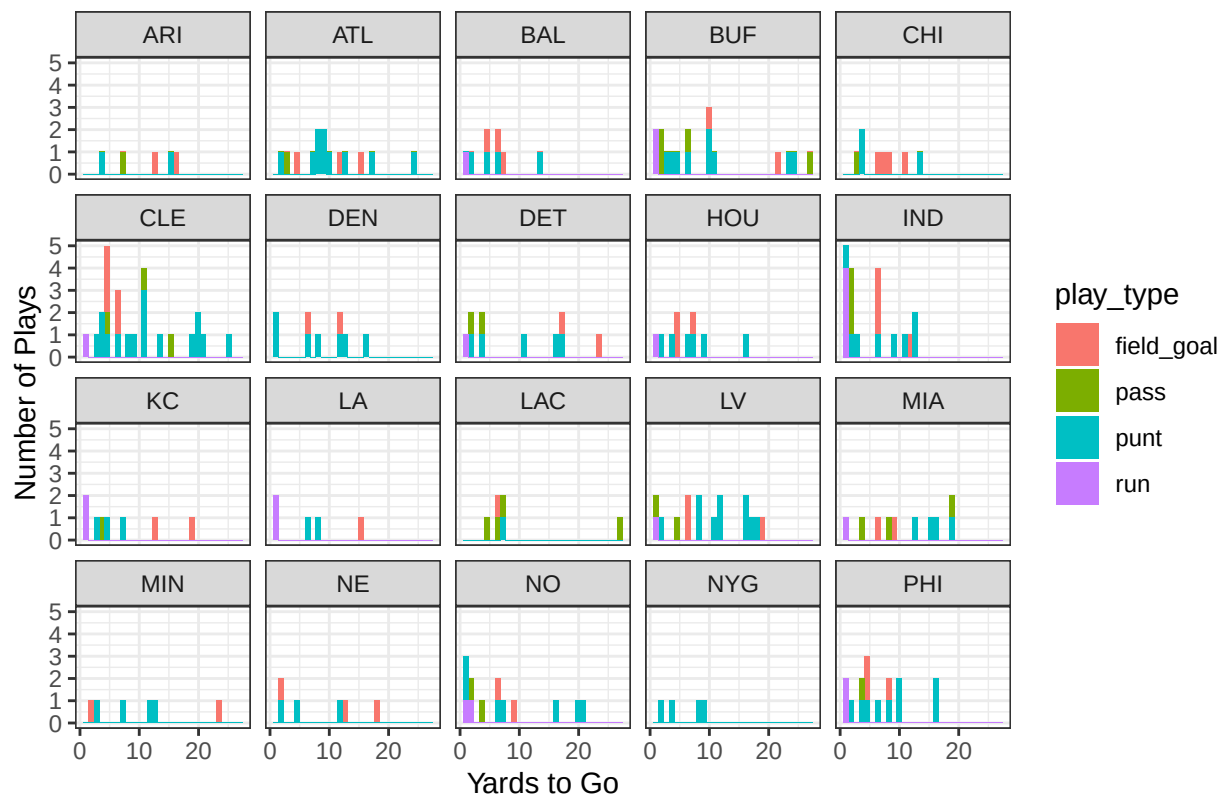


For teams on average, there doesn't appear to be much relation between seconds remaining and fourth down plays. However, the Broncos, Bills, and Falcons do appear to punt more earlier in the game and go for it more later in the game.

```
ggplot(fourth_down, aes(x = ydstogo, fill = play_type)) +
  geom_histogram() +
  facet_wrap(~ posteam) +
  labs(title = "4th-Down Decisions by Yards to Go (All Games)",
        x = "Yards to Go",
        y = "Number of Plays") +
  theme_bw()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

4th-Down Decisions by Yards to Go (All Games)



Interestingly, there doesn't appear to be much correlation between yards to a first down and fourth down action. The fourth down actions seem to be spread evenly across the distribution of yards to go. It appears to me that different teams have different risk tolerances for fourth down and that's consistent across different situations. For example, the Falcons and the Browns punt the ball a high percent of the team, meaning they're generally risk averse. This attitude holds despite the situation. Other teams like the Chargers or Chiefs don't have this same risk aversion and will choose to go for it even when they have a long way to go for the first down.

Question 2)

```
safeLeadData <- nflData %>%
  filter(game_id %in% games)

safeLeadData <- safeLeadData %>%
  mutate(
    lead = posteam_score - defteam_score,
    minutes_left = game_seconds_remaining / 60
  ) %>% filter(
    down == 1,
    ydstogo == 10,
    lead > minutes_left
  )

safeLeadData
```



```
## -- nflverse play by play data -----
## i Data updated: 2025-11-23 20:53:33 EST
## # A tibble: 59 x 374
##   play_id game_id      old_game_id home_team away_team season_type week posteam
##   <dbl> <chr>      <chr>      <chr>      <chr>      <chr>      <int> <chr>
## 1 3408 2025_02_PH~ 2025091411 KC        PHI        REG        2 PHI
## 2 3481 2025_02_PH~ 2025091411 KC        PHI        REG        2 PHI
## 3 3859 2025_02_PH~ 2025091411 KC        PHI        REG        2 PHI
## 4 3974 2025_02_PH~ 2025091411 KC        PHI        REG        2 PHI
## 5 3013 2025_05_LV~ 2025100503 IND        LV        REG        5 IND
## 6 3604 2025_05_LV~ 2025100503 IND        LV        REG        5 IND
## 7 3652 2025_05_LV~ 2025100503 IND        LV        REG        5 IND
## 8 3699 2025_05_LV~ 2025100503 IND        LV        REG        5 IND
## 9 4260 2025_06_AR~ 2025101203 IND        ARI        REG        6 IND
## 10 4311 2025_06_AR~ 2025101203 IND        ARI        REG        6 IND
## # i 49 more rows
## # i 366 more variables: posteam_type <chr>, defteam <chr>, side_of_field <chr>,
## # yardline_100 <dbl>, game_date <chr>, quarter_seconds_remaining <dbl>,
## # half_seconds_remaining <dbl>, game_seconds_remaining <dbl>,
## # game_half <chr>, quarter_end <dbl>, drive <dbl>, sp <dbl>, qtr <dbl>,
## # down <dbl>, goal_to_go <int>, time <chr>, yrdln <chr>, ydstogo <dbl>,
## # ydsnet <dbl>, desc <chr>, play_type <chr>, yards_gained <dbl>, ...
```

```
winners <- safeLeadData %>%
  group_by(game_id, home_team, away_team) %>%
  summarize(
    home_final = last(home_score),
    away_final = last(away_score),
    winning_team = ifelse(home_final > away_final, home_team, away_team)
  )
```

```
## `summarise()` has grouped output by 'game_id', 'home_team'. You can override
## using the `.groups` argument.
```

```
winners
```

```
## # A tibble: 14 x 6
## # Groups:   game_id, home_team [14]
##   game_id      home_team away_team home_final away_final winning_team
##   <chr>      <chr>      <chr>      <int>      <int> <chr>
## 1 2025_02_PHI_KC KC        PHI        17        20 PHI
## 2 2025_05_LV_IND IND        LV        40         6 IND
## 3 2025_06_ARI_IND IND        ARI        31        27 IND
## 4 2025_06_BUF_ATL ATL        BUF        24        14 ATL
## 5 2025_06_PHI_NYG NYG        PHI        34        17 NYG
## 6 2025_07_IND_LAC LAC        IND        24        38 IND
## 7 2025_07_MIA_CLE CLE        MIA        31         6 CLE
## 8 2025_07_NO_CHI CHI        NO        26        14 CHI
## 9 2025_08_CLE_NE NE        CLE        32        13 NE
## 10 2025_09_MIN_DET DET        MIN        24        27 MIN
## 11 2025_09_NO_LA LA        NO        34        10 LA
## 12 2025_10_LV_DEN DEN        LV        10         7 DEN
## 13 2025_11_BAL_CLE CLE        BAL        16        23 BAL
## 14 2025_12_BUF_HOU HOU        BUF        23        19 HOU
```

```

safeLeadResults <- safeLeadData %>%
  left_join(winners, by = "game_id") %>%
  mutate(safe_lead_success = posteam == winning_team)
safeLeadResults

## -- nflverse play by play data -----
## i Data updated: 2025-11-23 20:53:33 EST

## # A tibble: 59 x 380
##   play_id game_id old_game_id home_team.x away_team.x season_type week posteam
##   <dbl> <chr> <chr> <chr> <chr> <chr> <int> <chr>
## 1 3408 2025_0~ 2025091411 KC PHI REG 2 PHI
## 2 3481 2025_0~ 2025091411 KC PHI REG 2 PHI
## 3 3859 2025_0~ 2025091411 KC PHI REG 2 PHI
## 4 3974 2025_0~ 2025091411 KC PHI REG 2 PHI
## 5 3013 2025_0~ 2025100503 IND LV REG 5 IND
## 6 3604 2025_0~ 2025100503 IND LV REG 5 IND
## 7 3652 2025_0~ 2025100503 IND LV REG 5 IND
## 8 3699 2025_0~ 2025100503 IND LV REG 5 IND
## 9 4260 2025_0~ 2025101203 IND ARI REG 6 IND
## 10 4311 2025_0~ 2025101203 IND ARI REG 6 IND
## # i 49 more rows
## # i 372 more variables: posteam_type <chr>, defteam <chr>, side_of_field <chr>,
## # yardline_100 <dbl>, game_date <chr>, quarter_seconds_remaining <dbl>,
## # half_seconds_remaining <dbl>, game_seconds_remaining <dbl>,
## # game_half <chr>, quarter_end <dbl>, drive <dbl>, sp <dbl>, qtr <dbl>,
## # down <dbl>, goal_to_go <int>, time <chr>, yrdln <chr>, ydstogo <dbl>,
## # ydsnet <dbl>, desc <chr>, play_type <chr>, yards_gained <dbl>, ...

numSafeLeads <- nrow(safeLeadData)
numSuccessful <- sum(safeLeadResults$safe_lead_success)

numSafeLeads

## [1] 59

numSuccessful

## [1] 59

```

My 15 games consisted of 59 safe leads and every single one of those leads resulted in a win for the team with the safe lead.